PEAS Description and Task Environment Analysis

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Course Activity: Intelligent Agents, Task Environments (Section 2.2)

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Abstract

This report analyses eight everyday and extreme activities as task environments for intelligent agents. For each activity it gives a PEAS description (Performance measure, Environment, Actuators, Sensors) and classifies the environment on the dimensions of Section 2.2: observability, number of agents, determinism, episodic or sequential structure, dynamics, discreteness and knowledge of the rules.

The analysis shows that environments differ sharply in difficulty. Playing soccer is hard on all six factors scored here, while knitting a sweater is the simplest of the eight. Each environment is then matched to the agent design it calls for.

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1. Introduction

An intelligent agent is defined by its task environment. The PEAS framework describes that environment through four components:

- **Performance measure:** the criterion used to judge how well the agent is doing.
- **Environment:** the world the agent operates in.
- **Actuators:** the means by which the agent acts on the environment.
- **Sensors:** the means by which the agent perceives the environment.

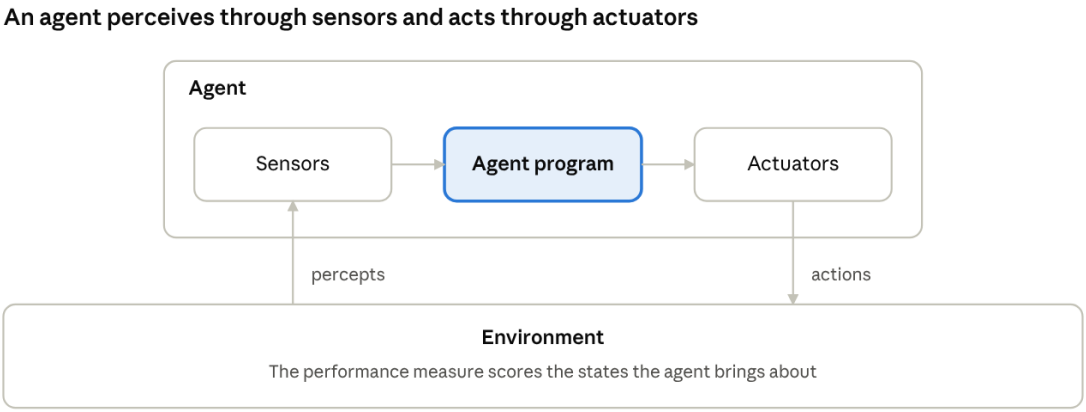
Each environment is then classified along the standard dimensions from Section 2.2:

Dimension	Possible values
Observability	Fully observable or Partially observable
Agents	Single agent or Multiagent (competitive or cooperative)
Determinism	Deterministic, Stochastic or Strategic
Episodic nature	Episodic or Sequential
Dynamics	Static, Semidynamic or Dynamic
State and time	Discrete or Continuous
Knowledge	Known or Unknown (rules of the environment)

1.1 Agents, Environments and Rationality

An agent is anything that perceives its environment through sensors and acts on that environment through actuators. A human player has eyes and ears for sensors and limbs for actuators; a robot has cameras and motors; a software agent has input data and output messages.

A rational agent chooses, for every possible percept sequence, the action expected to maximise its performance measure. That choice depends on four things: the performance measure, the agent's prior knowledge of the environment, the actions it can perform, and the percept sequence to date. Rationality is not omniscience: a rational agent can still fail because it could not know everything, but it never fails for lack of sensible reasoning about what it could perceive.



agent-environment loop · after Russell and Norvig, Section 2.1

Percepts flow in through the sensors, the agent program selects an action, and the actuators change the environment. The cycle then repeats, and the performance measure judges the whole sequence of environment states.

1.2 The PEAS Framework in Detail

PEAS is a checklist for specifying a task environment before any agent is designed. Writing it down forces the designer to say what success means, what the agent can sense, and what it can do.

Component	Question it answers	What to include
Performance measure	How is success judged?	Objective, measurable criteria such as goals scored, time taken, cost or safety
Environment	Where does the agent operate?	The physical or virtual setting, other agents, and conditions such as weather
Actuators	How can the agent change the world?	Every action the agent can perform
Sensors	How can the agent observe the world?	Every percept the agent can receive

A good performance measure rewards what the designer wants in the environment, not how the agent is supposed to behave. A vacuum cleaner rewarded for the amount of dirt collected, for example, can learn to dump dirt and collect it again. The same care applies to each activity in this report.

1.3 Properties of Task Environments in Detail

The seven properties below decide how hard an environment is and which agent design suits it. In each row the second column is the simpler case and the third is the harder one.

Property	Simpler case	Harder case	Why it matters
Observability	Fully observable: sensors reveal the complete relevant state	Partially observable: sensors are noisy, limited or incomplete	Partial observability forces the agent to keep an internal state or memory
Number of agents	Single agent	Multiagent, competitive or cooperative	Other agents must be predicted, and may need communication or strategy
Determinism	Deterministic: the next state is fixed by the current state and action	Stochastic: outcomes are uncertain; strategic: deterministic except for other agents	Uncertainty must be handled with probabilities or robust plans
Episodic or sequential	Episodic: each decision is independent of the last	Sequential: current choices affect future ones	Sequential tasks need lookahead and planning
Static or dynamic	Static: the world waits while the agent thinks	Dynamic: the world changes while the agent deliberates (semidynamic: only the score changes with time)	Dynamic worlds demand fast decisions
Discrete or continuous	Discrete: a finite set of states, percepts and actions	Continuous: values such as position and speed vary smoothly	Continuous control needs different methods from symbolic search
Known or unknown	Known: the agent knows the rules and outcomes	Unknown: the agent must learn how the environment works	Unknown environments require learning and exploration

The hardest case combines partial observability, several agents, stochastic outcomes, sequential decisions, a dynamic world, continuous values and unknown rules. Driving a taxi is the textbook example, and several of the activities below come close to it.

1.4 Scope and Method

Each activity is treated as a task for an agent, whether that agent is a person or a machine. The method is the same for all eight:

1. State the PEAS description of the task.
2. Classify the environment on the seven properties and justify each choice.
3. Describe percepts, actions and the challenges that follow.
4. Compare the eight activities side by side (Sections 3 and 4).
5. Match each environment to a suitable agent design (Section 5).

Some classifications are judgment calls. Used-book shopping can be single-agent or multiagent depending on whether auctions are involved, and a knitter may hold the pattern partly in memory. Where a choice is borderline, the report states the assumption it makes.

2. Activity Analysis

A) Playing Soccer

PEAS	Description
Performance measure	Goals scored versus conceded, match result, possession, fouls avoided
Environment	Pitch, ball, teammates, opponents, referee, weather
Actuators	Legs, feet, head, torso, voice (communication)
Sensors	Eyes, ears, touch and balance senses, proprioception

Properties: Partially observable (cannot see the whole pitch or others' intentions), multiagent (cooperative with teammates, competitive against opponents), stochastic, sequential, dynamic, continuous.

Scenario and rational behaviour

A soccer player, human or robot, must help the team score while preventing the opposing team from scoring. A rational player positions itself, passes, shoots and defends in the way expected to maximise the team's goal difference over the match, given what it can currently see of the pitch.

Percepts and actions

Percept	Typical action in response
Ball at the feet, defender approaching	Dribble past, pass or shoot
Teammate unmarked in open space	Pass to that player
Opponent with the ball near the goal	Close down, block or tackle
Referee whistle or signal	Stop, restart or accept the decision

Why each property holds

- **Partially observable:** a player cannot see behind, cannot see through other players and cannot know what opponents intend.
- **Multiagent:** teammates cooperate and opponents compete, so the environment is cooperative and competitive at once.
- **Stochastic:** bounces, deflections, wind and refereeing decisions make outcomes uncertain.
- **Sequential:** a pass now shapes the next several seconds and, eventually, the result.
- **Dynamic:** play continues while the player is still deciding, so slow deliberation loses the ball.
- **Continuous:** positions, speeds and angles take real values.
- **Known rules, unknown opponents:** the laws of the game are known, but the opposing tactics are not.

Suitable agent design

A model-based, utility-based agent with learning suits soccer. It keeps a world model of where the players and the ball are, estimates the value of each option (for example the chance that a shot scores), and learns opponents' habits over time. Fast reflexes, such as controlling a bouncing ball, run beneath slower strategic reasoning, such as set plays.

Main challenges

The agent must decide in real time on incomplete information, coordinate with teammates without perfect communication, and predict opponents who are trying to be unpredictable. Robot soccer competitions such as RoboCup use exactly this task as a research benchmark.

B) Exploring the Subsurface Oceans of Titan

PEAS	Description
Performance measure	Area explored, scientific data gathered, samples analysed, survival of the probe, energy efficiency
Environment	Titan's icy crust and subsurface liquid ocean, extreme cold, high pressure, unknown chemistry
Actuators	Thrusters, propellers, drilling or melting probe, robotic arm, radio transmitter
Sensors	Sonar, cameras, pressure, temperature and chemical sensors, accelerometers, communication receiver

Properties: Partially observable, single agent, stochastic, sequential, dynamic, continuous. The environment is largely unknown, so the agent must learn how it behaves.

Scenario and rational behaviour

Evidence gathered by the Cassini mission suggests that a liquid water ocean lies beneath Titan's icy crust, and no probe has yet entered it. A rational exploration agent would move through the ocean, measure it, collect samples and report its findings while keeping itself alive, since a lost probe cannot be replaced.

Percepts and actions

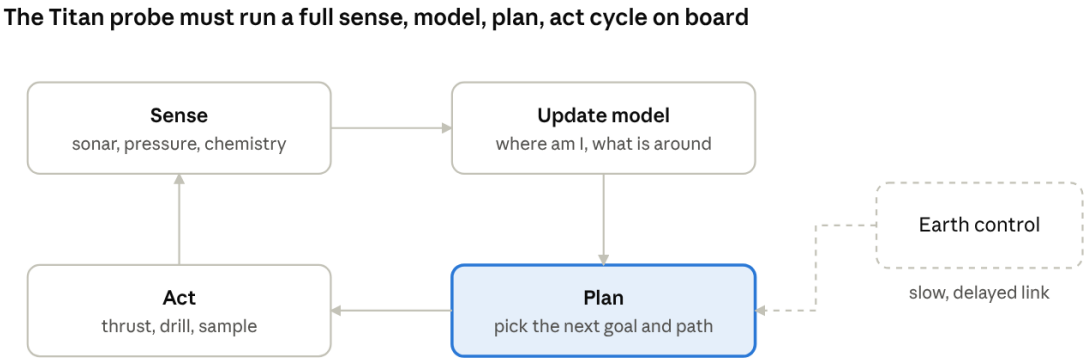
Percept	Typical action in response
Pressure rising sharply with depth	Hold depth or change course
Sonar returns an unexpected obstacle	Steer around it
Battery running low	Conserve energy and transmit the data collected
Unusual chemical signature	Slow down and collect a sample

Why each property holds

- **Partially observable:** sonar and cameras reach only a short distance, and most of the ocean is hidden at any moment.
- **Single agent:** one probe operates alone. A swarm of cooperating probes would make the task multiagent.
- **Stochastic:** unmodelled currents, ice movement and sensor noise make the result of an action uncertain.
- **Sequential:** decisions about depth, energy and sampling sites build on one another.
- **Dynamic:** currents and ice keep changing while the probe deliberates.
- **Continuous:** position, depth, pressure and temperature vary smoothly.
- **Unknown:** the physics and chemistry of the ocean are largely unknown, so the agent must learn how it behaves.

Suitable agent design

The probe needs an autonomous, goal-based and learning agent. It keeps a model of the ocean, updates that model from its sensors and plans actions toward scientific goals. Simple fail-safe reflexes protect it from immediate hazards, and the link to Earth carries only high-level goals and summaries.



autonomy loop of the Titan probe, 4 steps plus a delayed Earth link

The loop runs entirely on board. Earth supplies goals over a slow link, because waiting for a command at every step would be too slow.

Main challenges

The one-way signal delay between Earth and Saturn is more than an hour, so remote control is impractical and the agent must act on its own. Extreme cold, high pressure, a limited power budget and the absence of a second chance add to the difficulty.

C) Shopping for Used AI Books on the Internet

PEAS	Description
Performance measure	Low price, good book condition, relevance to the topic, reliable seller, delivery time
Environment	Websites, search engines, seller listings, payment systems, shipping services
Actuators	Typing queries, clicking links, filling forms, submitting payment
Sensors	Web page content, listing descriptions, images, ratings and reviews

Properties: Partially observable (book condition and seller honesty are uncertain), single agent (multiagent if auctions or competing buyers are involved), partly stochastic (stock and prices change), sequential, semidynamic to dynamic, discrete.

Scenario and rational behaviour

A shopping agent is asked to buy used books on artificial intelligence at a good price and in acceptable condition. A rational agent searches several sites, compares listings, checks seller reliability and buys when the total cost, quality and delivery time are better than the alternatives it has seen.

Percepts and actions

Percept	Typical action in response
Search results page	Refine the keywords or open a listing
Listing with price and condition notes	Compare against the budget and other listings
Seller rating and reviews	Trust the seller or skip the listing
Shipping cost shown at checkout	Recompute the total and decide whether to buy

Why each property holds

- **Partially observable:** the agent sees descriptions and photos only. The true condition of the book, the seller's honesty and the stock behind a listing are hidden.
- **Single agent:** fixed-price listings involve no competitor. Auctions or limited stock would add competing buyers and make the task multiagent.
- **Partly stochastic:** opening a page is predictable, but prices change, listings sell out and delivery times vary.
- **Sequential:** each search narrows the next one, and a purchase changes what is still needed.
- **Semidynamic to dynamic:** the web changes while the agent browses, but only slowly compared with a game.
- **Discrete:** pages, links, prices and quantities form a finite, countable set of states and actions.
- **Known interface:** the way a web shop works is familiar, although the practices of individual sellers are not.

Suitable agent design

A goal-based, utility-based agent with memory fits this task. It remembers the listings already seen, scores each one on price, condition, seller trust and delivery time, and buys the highest-scoring option. A learning component can improve its ability to spot unreliable sellers and misleading descriptions.

Main challenges

The main difficulty is uncertainty about quality: the agent must act on incomplete and sometimes misleading information. Search results are also unreliable as a measure of what is available, and prices shift while the agent compares them. The performance measure should include shipping and returns, because the lowest listed price is often not the lowest total cost.

D) Playing a Tennis Match

PEAS	Description
Performance measure	Points, games and sets won, unforced errors, winning the match
Environment	Court, net, ball, opponent, umpire, wind and weather
Actuators	Arms, racket, legs, body movement
Sensors	Eyes (ball and opponent position), ears, proprioception

Properties: Fully observable (ball, court and opponent are visible), multiagent and competitive, stochastic, sequential, dynamic, continuous.

Scenario and rational behaviour

A tennis player competes against one opponent to win points, games and sets. A rational player chooses each shot to maximise the chance of winning the point, while also managing energy over a match that can last for hours.

Percepts and actions

Percept	Typical action in response
Ball leaving the opponent's racket	Predict its path and move to meet it
Opponent out of position	Hit to the open side of the court
Short, high ball	Move forward and attack with a smash or volley
Score 30-40 on own serve	Choose a safer, higher-percentage serve

Why each property holds

- **Fully observable:** the court, the ball and the opponent's position are visible to both players at all times. The opponent's intentions and the spin on the ball are not directly visible, so observability here refers to the physical state of the game.
- **Multiagent and competitive:** one player's gain is the other's loss, which makes the opponent the central part of the environment.
- **Stochastic:** wind, net cords, line calls and small errors in timing mean that the same shot does not always give the same result.
- **Sequential:** shots within a rally depend on each other, and points accumulate into games and sets. Each point is partly independent, but fatigue and momentum carry over.
- **Dynamic:** the ball keeps moving and the opponent keeps acting while the player decides.
- **Continuous:** ball position, speed, spin and player movement take real values.
- **Known rules:** the rules of tennis are fixed and fully known.

Suitable agent design

A model-based, utility-based agent with opponent modelling suits tennis. Reflexes handle the fast, low-level work of reaching and returning the ball. A strategic layer chooses shot placement, learns the opponent's weaknesses and adapts as the match develops.

Main challenges

Decisions must be made within fractions of a second while the opponent is adapting at the same time. Because the opponent is trying to defeat the agent's strategy, a fixed pattern of play is soon exploited. The agent must therefore keep some unpredictability in its choices, which is a standard result from game theory.

E) Practicing Tennis Against a Wall

PEAS	Description
Performance measure	Number of consecutive returns, accuracy, consistency of shots
Environment	Practice wall, ball, ground surface, wind
Actuators	Arms, racket, legs, body movement
Sensors	Eyes, ears, proprioception

Properties: Fully observable, single agent (the wall is not an agent), stochastic (spin and bounce vary), sequential, dynamic, continuous.

Scenario and rational behaviour

A player hits a ball against a wall and returns it as it rebounds. There is no opponent, so the goal is to improve technique and consistency. A rational player varies the shots, the speed and the position, and aims to keep long, accurate rallies going.

Percepts and actions

Percept	Typical action in response
Ball rebounding low and fast	Step back and take it after the bounce
Ball returning to the left side	Move left and use a backhand
Previous shot hit the top of the wall	Lower the racket path on the next shot
Ball slowing after a bad bounce	Move in and adjust the swing

Why each property holds

- **Fully observable:** the wall, the ball and the player's own position are always visible.

- **Single agent:** the wall is a passive object that returns the ball according to physics and does not choose actions, so it is not an agent.
- **Stochastic:** spin, surface irregularities and slight variations in contact make each rebound slightly different.
- **Sequential:** each shot decides the position and timing of the next, so one mistake affects the rest of the rally.
- **Dynamic:** the ball keeps moving while the player decides.
- **Continuous:** speed, angle, spin and position all take real values.
- **Known:** the environment's rules are simple, although the exact bounce of each ball must be learned.

Comparison with a match

Removing the opponent changes the task in three ways. The environment becomes single-agent, it becomes more predictable because the wall does not adapt, and the goal shifts from beating a rival to improving the agent itself. The same player therefore faces an easier environment than in a match, which is why wall practice is used for training.

Suitable agent design

A learning agent built on a model-based reflex agent is the natural choice. The reflexes handle the returns, and a learning element adjusts the swing from feedback on each shot, in the manner of reinforcement learning. Because the environment does not change its behaviour, the agent can improve steadily with repetition.

Main challenges

The difficulty is precision rather than strategy: the agent must control a continuous, fast-moving system with small timing errors. Fatigue and boredom also degrade performance during long practice sessions.

F) Performing a High Jump

PEAS	Description
Performance measure	Height cleared, successful clearance of the bar, form and safety
Environment	Run-up area, bar and uprights, landing mat, wind
Actuators	Legs, arms, torso for run-up, take-off, bar clearance and landing

PEAS	Description
Sensors	Eyes, balance and proprioception, sense of timing

Properties: Fully observable, single agent, stochastic (small execution variations affect the result), episodic (each jump is largely independent), static (the bar does not move), continuous.

Scenario and rational behaviour

A high jumper runs up, takes off, clears a horizontal bar and lands on a mat. A rational jumper sets the bar at the highest height that gives a good chance of clearing, and plans the run-up, take-off and body position to achieve it.

Percepts and actions

Percept	Typical action in response
Bar height set by the officials	Choose the starting height and the number of attempts to use
Foot placement at the start of the curve	Adjust the stride length
Body angle at take-off	Rotate over the bar
Bar wobbling after contact	Land and record a clearance or a fault

Why each property holds

- **Fully observable:** the bar, the uprights, the mat and the jumper's own body position are directly visible or felt.
- **Single agent:** the jumper competes against the bar, and other athletes do not interact with the attempt.
- **Stochastic:** tiny differences in speed, angle and timing change whether the bar stays up, so the same plan does not always give the same outcome.
- **Episodic:** each attempt is largely independent of the one before it. Fatigue and the order of heights add a weak link, so this classification is approximate.
- **Static:** the bar and the mat do not move while the jumper is deciding. Wind is a small exception.
- **Continuous:** speed, angle, height and body position take real values.
- **Known:** the rules of the event and the physics of jumping are well understood.

Suitable agent design

A simple model-based reflex agent with learned parameters is enough. Because attempts are almost independent, the agent does not need deep planning. It improves through repetition by adjusting its run-up, take-off point and body angle from the result of each attempt.

Main challenges

The task demands extremely precise timing and body control in a very short time. A small error in the take-off point can decide the attempt, and the risk of injury limits how aggressively the agent can explore. Choosing which heights to attempt is the only part of the task that involves longer-term strategy.

G) Knitting a Sweater

PEAS	Description
Performance measure	Correct pattern and size, even stitches, no defects, completion time
Environment	Yarn, needles, pattern instructions, work surface
Actuators	Fingers and hands (looping, pulling, crossing stitches)
Sensors	Eyes, touch, reading the pattern

Properties: Fully observable (the work in progress is visible, though a pattern plan may be partly remembered), single agent, deterministic, sequential (each stitch depends on earlier ones), static, discrete.

Scenario and rational behaviour

A knitter follows a pattern to build a sweater stitch by stitch and row by row. A rational knitter works steadily from the pattern, checks the work against the target size and stitch count, and corrects errors before they spread.

Percepts and actions

Percept	Typical action in response
Current row number and pattern instruction	Knit, purl, increase or decrease as written
Yarn too tight or too loose	Adjust the tension
Dropped or twisted stitch	Pick it up or undo back to the error

Percept	Typical action in response
Row count reaches the sleeve length	Cast off and start the next piece

Why each property holds

- **Fully observable:** the work in progress, the needles and the written pattern are all in plain view. The assumption is that the pattern is read as the row is worked, so nothing important is hidden in memory.
- **Single agent:** one person works alone, and no other agent changes the knitting.
- **Deterministic:** given an action, the stitch results as intended. Slips and dropped stitches are rare exceptions that can be treated as faults.
- **Sequential:** each stitch depends on those before it, and an early mistake carries into every row above it.
- **Static:** the knitting does not change unless the knitter acts, so there is no time pressure from the environment.
- **Discrete:** stitches, rows and instructions are countable, with a finite set of operations.
- **Known:** the pattern and the rules of the craft are fully specified.

Suitable agent design

A goal-based planning agent that follows a plan is sufficient. The pattern is the plan, a model of the current row and stitch count tracks progress, and a simple check compares the work with the pattern. Because the environment is so predictable, little learning is needed beyond improving tension and speed.

Main challenges

The task is long rather than hard: a sweater needs thousands of correct stitches, so the main risk is a small error early on that is discovered much later. Error detection is the most valuable skill, because undoing many rows is costly. Among the eight activities this is the simplest environment, and a robot knitter would need to master fine manipulation of soft, deformable yarn.

H) Bidding on an Item at an Auction

PEAS	Description
Performance measure	Winning the item at a good price (value obtained minus price paid), staying within budget

PEAS	Description
Environment	Auction house or platform, item, other bidders, auctioneer, bidding rules
Actuators	Placing bids, raising a paddle, clicking a bid button, withdrawing
Sensors	Current bid, auctioneer announcements, item description, observation of rival bidders

Properties: Partially observable (other bidders' valuations are hidden), multiagent and competitive, strategic or stochastic, sequential, dynamic (the price changes in real time), discrete.

Scenario and rational behaviour

A bidder wants to win an item without paying more than it is worth. A rational bidder decides a maximum price from the item's private value to it, bids only while the price stays below that value, and stops when it is exceeded. The aim is to maximise the surplus: the value of the item minus the price paid.

Percepts and actions

Percept	Typical action in response
Current highest bid	Raise the bid or hold
Auctioneer announces a closing call	Place a final bid or withdraw
Rival bidders keep raising quickly	Infer their valuation is high and consider stopping
Price reaches the personal maximum	Withdraw from the auction

Why each property holds

- **Partially observable:** the bidder sees the current price and the item, but not the other bidders' maximum prices or their reasons for bidding.
- **Multiagent and competitive:** each bidder's success depends on what the others do, and only one can win.
- **Strategic:** the process is governed by fixed rules, and the uncertainty comes mainly from the other agents' choices. From the bidder's point of view this appears stochastic.
- **Sequential:** bids come in rounds, and each bid changes the price and what rivals do next.
- **Dynamic:** the price rises in real time while the bidder is deciding, and the auction can close at any moment.

- **Discrete:** bids move in fixed increments and the possible actions form a finite set.
- **Known rules, unknown bidders:** the format of the auction is known, whereas the valuations and tactics of the other bidders are not.

Suitable agent design

A utility-based agent with game-theoretic reasoning is the best fit. It compares the item's value with the price at each step, estimates the other bidders' behaviour from the bids it observes, and chooses actions that maximise expected surplus. A learning component can recognise patterns such as late bidding by a particular rival.

Main challenges

The central risk is the winner's curse: the winner is often the bidder who most overestimated the item's value, so a rational agent shades its bids to allow for this. Last-second bidding in online auctions adds a timing problem, and the agent must also resist emotional or competitive overbidding, which a human bidder may struggle with and a well-designed agent avoids.

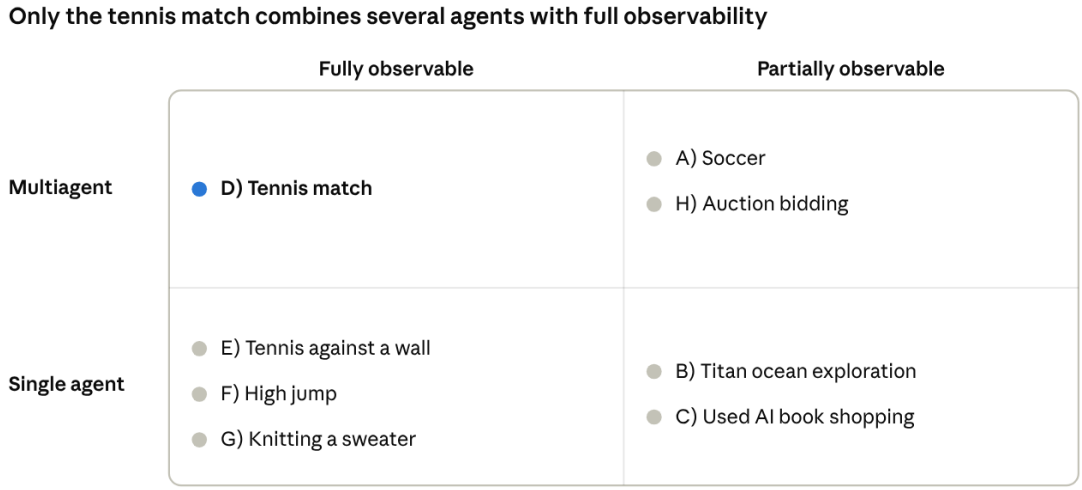
3. Summary Comparison

Activity	Observable	Agents	Deterministic	Episodic	Static	Discrete
A) Soccer	Partial	Multi	Stochastic	Sequential	Dynamic	Continuous
B) Titan ocean exploration	Partial	Single	Stochastic	Sequential	Dynamic	Continuous
C) Used AI book shopping	Partial	Single	Partly stochastic	Sequential	Semidynamic	Discrete
D) Tennis match	Full	Multi	Stochastic	Sequential	Dynamic	Continuous
E) Tennis against a wall	Full	Single	Stochastic	Sequential	Dynamic	Continuous
F) High jump	Full	Single	Stochastic	Episodic	Static	Continuous
G) Knitting a sweater	Full	Single	Deterministic	Sequential	Static	Discrete

Activity	Observable	Agents	Deterministic	Episodic	Static	Discrete
H) Auction bidding	Partial	Multi	Strategic	Sequential	Dynamic	Discrete

3.1 Placing the Activities on Two Axes

Observability and the number of agents together separate the eight activities into four groups.

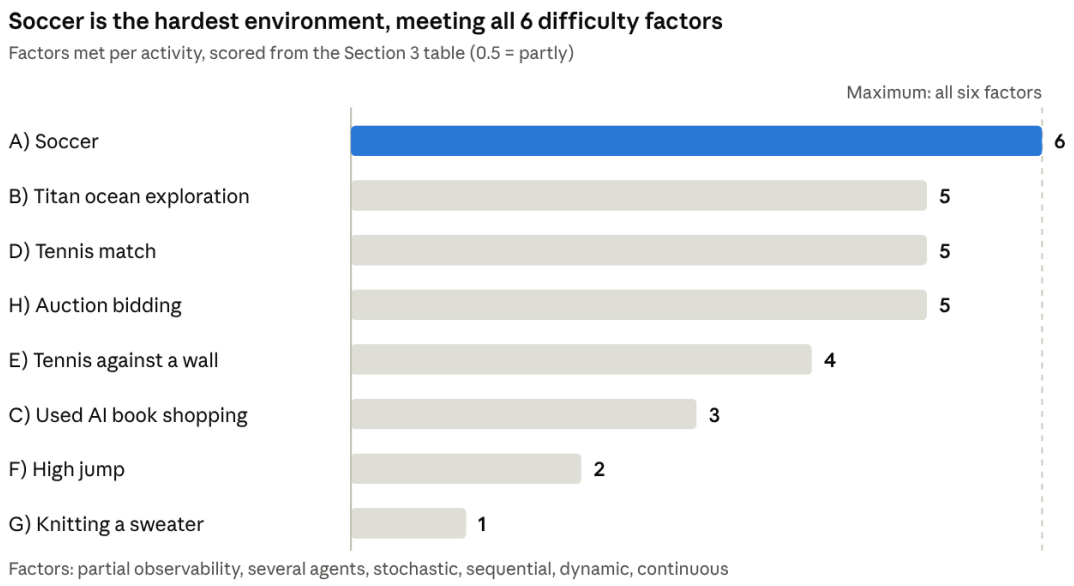


two-axis placement of the 8 activities, observability by number of agents

The tennis match is the only activity that has several agents yet lets the player see the whole physical state. Soccer and auctions add partial observability to the presence of other agents, and they are the cases where modelling other agents matters most.

3.2 Environment Difficulty by Activity

The chart counts how many of six difficulty factors apply to each activity: partial observability, several agents, stochastic outcomes, sequential decisions, a dynamic world and continuous values. A factor scores 1 when it fully applies and 0.5 when it applies partly, as with the semidynamic, partly stochastic shopping task. The known or unknown dimension is left out because the rules are known in every case except Titan.



scored from the Section 3 classification table, 6 factors, 8 activities

Soccer meets every factor. Titan, the tennis match and auction bidding each score 5, but for different reasons: Titan is difficult because of what the agent cannot see, the match because of its opponent, and the auction because of hidden valuations. Knitting scores 1 because only the sequential structure applies. The scores are an aid to comparison and not a precise measurement, since the factors are not equally hard.

4. Comparative Discussion

Comparing the eight activities property by property shows which features of an environment drive difficulty and which activities share a design problem.

4.1 Property by Property

Property	Harder side	Simpler side
Observability	A) Soccer, B) Titan, C) Book shopping, H) Auction	D) Tennis match, E) Wall practice, F) High jump, G) Knitting
Number of agents	A) Soccer, D) Tennis match, H) Auction	B) Titan, C) Book shopping, E) Wall practice, F) High jump, G) Knitting
Determinism	A, B, D, E, F stochastic; H strategic; C partly stochastic	G) Knitting is deterministic

Property	Harder side	Simpler side
Episodic or sequential	All activities except F are sequential	F) High jump is episodic
Static or dynamic	A, B, D, E, H dynamic; C semidynamic	F) High jump and G) Knitting are static
Discrete or continuous	A, B, D, E, F continuous	C) Book shopping, G) Knitting, H) Auction are discrete

Observability. Four activities hide part of the state from the agent. In soccer and the auction the hidden part is mostly other agents' intentions and valuations. In the Titan ocean it is the physical surroundings, and in book shopping it is the true quality of the item. The common consequence is that the agent must keep an internal model and reason about what it cannot see.

Number of agents. Three activities are multiagent, and they are the ones in which the opponent is the main source of difficulty. Soccer is both cooperative and competitive, tennis and auctions are purely competitive. Book shopping is single-agent only under the assumption of fixed-price listings.

Determinism. Only knitting is close to deterministic. Tennis, soccer and the high jump are stochastic because of tiny physical variations, and Titan is stochastic because the environment is poorly modelled. The auction is strategic: its rules are fixed, but the outcome depends on other agents' choices.

Episodic or sequential. Almost every activity is sequential, which is typical of real tasks. The high jump is the nearest to episodic, because each attempt is largely separate, although even there fatigue and the ordering of heights create a weak link.

Static or dynamic. The high jump and knitting wait for the agent, so it can take as long as it needs. In the remaining activities the world keeps moving, which forces fast decisions and, in practice, a reflex layer beneath slower reasoning.

Discrete or continuous. Physical activities that involve movement are continuous, while activities defined by menus, bids or stitches are discrete. Continuous tasks lean on control methods, and discrete tasks on search and planning.

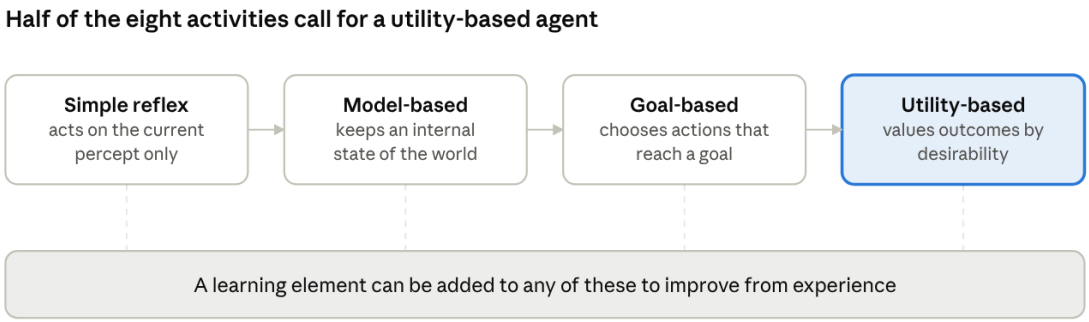
4.2 Patterns Across the Eight Activities

1. **Removing the opponent simplifies the task.** The tennis match and wall practice differ mainly in whether another agent is present. Wall practice is single-agent and more predictable, which is why it works as training.
2. **Time pressure shapes the design.** Dynamic, continuous environments such as soccer and tennis call for a fast reflex layer, while static environments such as knitting allow slow, careful planning.
3. **Hidden information changes the question.** In partially observable tasks such as the auction and shopping, the agent's most important job is to estimate what it cannot see.
4. **A classification depends on assumptions.** Shopping becomes multiagent when auctions are involved, and knitting is only fully observable if the pattern is read as the work proceeds. A PEAS analysis is only as precise as the assumptions that are stated with it.
5. **Unknown rules are rare but serious.** Titan is the only activity in which the agent does not know how the environment works, and that is why it needs a learning component and caution when acting.

5. Agent Design Implications

The properties of an environment determine which agent design is likely to work. The classification in Section 3 therefore leads directly to a design choice for each activity.

5.1 The Agent Types



agent types in order of added capability, with a learning element

A **simple reflex agent** maps the current percept directly to an action and works only when the environment is fully observable and the right action depends on the present moment. A **model-based agent** adds an internal state, so that it can cope with partial observability by tracking what it cannot see. A **goal-based agent** compares possible action sequences against a goal and plans, which suits sequential

tasks. A **utility-based agent** goes further by scoring outcomes, which is needed when goals conflict or outcomes are uncertain. A **learning element** can be added to any type so that the agent improves with experience, and it is essential when the environment is unknown.

5.2 Recommended Design for Each Activity

Activity	Recommended agent	Reason
A) Soccer	Model-based, utility-based, with learning	Partial observability, opponents and real-time decisions
B) Titan exploration	Goal-based with learning, plus reflex safety layer	Unknown, partially observable environment and a delayed link to Earth
C) Used AI book shopping	Goal-based or utility-based, with memory	Hidden quality and several criteria to balance
D) Tennis match	Model-based, utility-based, with opponent model	A competing agent and very fast play
E) Wall practice	Model-based reflex agent with learning	A predictable, single-agent task that improves with repetition
F) High jump	Model-based reflex agent with learned parameters	Near-episodic task that depends on precise control
G) Knitting	Goal-based plan-following agent	Deterministic, static and fully specified by the pattern
H) Auction bidding	Utility-based, game-theoretic	Hidden valuations and competing bidders

5.3 Design Rules That Follow

- **Partial observability calls for internal state.** Any activity marked partially observable needs memory or a model of the world.
- **Multiple agents call for modelling the others.** Soccer, tennis and the auction reward an agent that predicts and adapts to its competitors.

- **Dynamic, continuous worlds call for a fast layer.** Reflexes should run beneath slower planning so that the agent can react in time.
- **Uncertain outcomes call for utilities.** When results are stochastic, comparing expected value is more reliable than aiming at a single goal.
- **Unknown environments call for learning and caution.** Titan is the clearest case: the agent must learn as it goes, without risking the mission.

6. Challenges and Limitations

The analysis rests on simplifying assumptions, and several of its results should be read with those assumptions in mind.

6.1 Limits of the Classification

- **Boundaries between categories are blurred.** Whether a task is static or dynamic, or episodic or sequential, can depend on the time scale chosen. A single tennis point is nearly episodic, while a whole match is clearly sequential.
- **The agent's identity matters.** A human knitter and a robot knitter face different sensors and actuators, so the same task can be classified differently for each.
- **Assumptions are part of the answer.** The treatment of shopping as single-agent, and of knitting as fully observable, depends on stated assumptions that a different reader could reasonably change.
- **Opponents are hard to classify.** In multiagent settings, an opponent's behaviour can look stochastic to the agent even when it follows a deterministic strategy, which is why the auction is described as strategic.

6.2 Limits of the Difficulty Score

The difficulty chart in Section 3.2 counts factors and treats them as equal. In practice a single severe factor can outweigh several mild ones: the unknown rules of the Titan ocean may matter more than any other feature of that task, yet they are excluded from the score. The score is therefore a convenient summary for comparison and not a measurement of how hard an agent would find each task.

6.3 Practical Lessons

1. **Specify the performance measure carefully.** A poorly chosen measure, such as the lowest price alone in book shopping, leads to agents that satisfy the measure and miss the purpose.

2. **Design for the hardest property first.** In each activity one property dominates the design: the opponent in tennis, hidden valuations in the auction, the unknown ocean in the Titan mission.
3. **Test in simulation before the real environment.** Where mistakes are costly, as on Titan or in a high jump, simulation allows safe learning before any real attempt.
4. **Expect the classification to change.** If an agent gains better sensors or an opponent changes tactics, the environment changes class and the design must be revisited.

6.4 Possible Extensions

The same analysis could be applied to further activities, such as driving a taxi, playing chess or diagnosing a patient. It could also be extended with a finer scale for each property, or with measured data from simulated agents, to replace the simple factor count with an empirical measure of difficulty.

7. Conclusion

The eight activities show that task environments vary widely. Competitive team games and auctions demand reasoning about other agents, while knitting and the high jump are single-agent and largely static. Exploring Titan is the hardest case because the environment is partially observable, dynamic and largely unknown. The environment type determines the agent design: simple, fully observable and deterministic tasks suit simple reflex or planning agents, whereas partially observable, multiagent and stochastic tasks call for agents that maintain internal state, reason about uncertainty and learn.

7.1 Key Findings

- Soccer is the hardest of the eight environments, meeting all six difficulty factors.
- Knitting a sweater is the simplest, because only its sequential structure applies.
- Four activities (soccer, Titan, book shopping and the auction) are partially observable, and three (soccer, the tennis match and the auction) are multiagent.
- Only the Titan mission faces an environment whose rules are largely unknown.
- Half of the activities (soccer, book shopping, the tennis match and the auction) call for a utility-based agent.

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